

Which chart when

Match the chart to what you want the data to show

Comparison — values across categories

Bar chart	compare quantities across categories
Grouped bar	sub-categories within groups
Stacked bar	part-to-whole across categories
Lollipop	cleaner than bar when many items

Trend — change over time

Line chart	continuous trend, one or few series
Area chart	trend with volume emphasis
Stacked area	parts contributing to a total

Distribution — spread of one variable

Histogram	frequency of one numeric variable
Box plot	median, quartiles, outliers, groups
Violin plot	shape and density across groups
KDE / density	smoothed distribution curve

Relationship — between variables

Scatter plot	correlation between two numerics
Bubble chart	three variables: x, y, size
Heatmap	correlation matrix or intensity
Pair plot	pairwise relations for EDA

Composition — part-to-whole

Stacked bar	prefer over pie for comparing groups
Treemap	hierarchical, many categories
Pie / donut	proportions, max 5-6 slices, sparingly

Ranking

Sorted bar	ordered compare
Bump chart	rank over time

Spatial / geographic

Choropleth	values by region
Symbol map	points on map

Quick rules

Time on x-axis → line or area	Avoid: pie with 6+ slices
Comparing categories → bar	Avoid: dual y-axes
One variable's spread → histogram or box plot	Avoid: 3D charts
Two variables' relationship → scatter	Avoid: truncated bar y-axis
Parts of a whole → stacked bar over pie	Avoid: line for non-continuous x